OO-P-0040D (GSA-FSS)
June 15, 1977
SUPERSEDING
Int. Fed. Spec. OO-P-0040C(GSA-FSS)
May 20, 1968

INTERIM FEDERAL SPECIFICATION

PAD, FLOOR POLISHING MACHINE: AND HOLDER, FLOOR POLISHING MACHINE PAD

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers floor maintenance pads used for polishing, buffing, scrubbing, and stripping floors and are suitable for mounting on power driven, rotary disc type floor maintenance machines in the range of 150 to 350 r.p.m.'s.

1.2 Classification.

1.2.1 Types and classes. Floor maintenance pads and floor pad holders shall be of the following types and classes as specified (see 6.2). Note: The colors below are only to indicate the most common colors in use by industry for that particular class pad. Procurement will be made by type and class only.

Type I - Thick floor pads, nylon (11/16 inch min. thick).

Class 1 - Buffing (Tan).
Class 2 - Scrubbing (Green).
Class 3 - Stripping (Black).
Class 4 - Polishing (White).

Type II - Thin floor pads, nylon (1/4 inch min. thick).

Class 1 - Buffing (Tan).
Class 2 - Scrubbing (Green).
Class 3 - Stripping (Black).

Type III - Floor pad holders.

Class 2 - Base plate and retaining collar design.
Class 3 - Insta-lok type material driving face.
Class 4 - Velcro type material driving face.

Type IV - Thick floor pads, curled hair (1 inch min. thick).

Class 1 - Buffing (Orange).
Class 2 - Scrubbing (Greenish Yellow).
Class 3 - Stripping (Green).
Class 4 - Polishing (Yellow).

Type V - Thin buffing pad, nylon or polyester (3/8 inch min. thick).

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Type VI - Thick spray pads, polyester (7/8 inch min. thick).

Class 1 - Spray Buffing (Red).
Class 2 - Spray Scrubbing (Blue).
Class 3 - Spray Stripping (Brown).
Class 4 - Spray Polishing (White).

Type VII - Thick floor pads, polyester and nylon blend (3/4 inch min. thick).

Class 1 - Buffing (Red).
Class 2 - Scrubbing (Blue).
Class 3 - Stripping (Black).
Class 4 - Polishing (White).

1.2.2 Sizes.

1.2.2.1 Types I, II, IV, V, VI, and VII pads and type III classes 3 and 4 floor pad holders shall be of the following sizes as specified (see 6.2):

Size 4	- 4 inch diameter.	Size 17	- 17 inch diameter.
Size 10	- 10 inch diameter.	Size 18	- 18 inch diameter.
Size 11	- 11 inch diameter.	Size 19	- 19 inch diameter.
Size 12	- 12 inch diameter.	Size 20	- 20 inch diameter.
Size 13	- 13 inch diameter.	Size 21	- 21 inch diameter.
Size 14	- 14 inch diameter.	Size 22	- 22 inch diameter.
Size 15	- 15 inch diameter.	Size 23	- 23 inch diameter.
Size 16	- 16 inch diameter.	Size 24	- 24 inch diameter.

1.2.2.2 Type III, class 2 floor pad holders shall be for use with pads up to 4-1/2 inches maximum diameter center hole.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

P-W-155 - Wax, Floor, Water-Emulsion.
OO-P-570 - Polisher, Floor, Electric (Commercial Type).
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The materials used in the manufacture of floor maintenance pads shall be either curled hair, nylon, polyester or a blend of nylon and polyester and suitable for the intended use in accordance with good commercial practice. Abrasive particles shall be made of either natural or synthetic abrasives. Bonding adhesives shall be of good quality. The materials used shall not support the growth of bacteria and must be of such denier, density, strength, quality, and workmanship to produce a cohesive pad, uniform in structure and reversible with equal performance on each side. The pads shall withstand repeated washings in strong detergent-ammonia (stripping) solutions for removal of accumulated wax, dirt, etc., and the matrix shall retain its original condition after rinsing with water and drying. Also, the pads shall not disintegrate in water, detergent solutions, oil, wax, or other material normally encountered in floor maintenance operations. Recycled materials shall be used where practicable.

3.2 Design. All floor pads and floor pad holders shall be suitably designed and fabricated for use with rotary disc electric floor polishers covered by OO-P-570 or other machines having driving heads similar to those of the above floor polishers. The floor pads shall be resilient so as to contact the floor over their entire area while in use. Thick style pads 10 inches in diameter and larger, shall be provided with a scored or removable center hole 3-1/2 inches plus or minus 1/8 inch in diameter. The sizes of the pads shall be in accordance with 1.2.2.1 as to the sizes specified within a tolerance of plus or minus 1/8 inch on the diameter. The center hole shall be concentric with the outside diameter within 1/8 inch, total runout. The pads shall be uniformily impregnated with abrasive particles as specified in 3.1. The abrasive content of the thick pads shall be uniform, within, plus or minus 15 percent throughout a cross sectional thickness of the pad when tested in accordance with 4.4.4.1.

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3.3 Detail requirements.

3.3.1 Type I floor pads. Type I floor pads, classified as thick nylon floor pads, shall be in accordance with table II when tested in accordance with 4.4.3 and 4.4.3.1. The pads shall be made of a one piece, unified mat. The materials of the pads shall be formed or bonded into a solid disc and a center hole shall be provided as specified in 3.2. Pads of spirally wound ribbons of interlocking fibers are not acceptable.

3.3.1.1 Class 1. Type I, class 1 buffing pads shall be suitable for polishing and buffing of various types of floors such as ceramic, vinyl, rubber, or asphalt tiles and marble or terrazzo. The abrasive content for buffer pads shall be no more than 35 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 200 mesh screen when tested in accordance with 4.4.4.2. A minimum of 99 percent of the remaining particles shall pass through a 270 mesh screen. The buffing pads, when tested in accordance with 4.4.5, shall remove no less than 0.00 g or no more than 0.50 g of weight when averaged from the substrate material. The buffing pads shall be capable of producing a glass reading of not less than 30 percent in the test tile when tested in accordance with 4.4.6.

3.3.1.2 Class 2. Type I, class 2 scrubbing pads shall be suitable for cleaning and scrubbing the type floors specified in 3.3.1.1. The abrasive content for scrubbing pads shall be no more than 60 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 150 mesh screen when tested in accordance with 4.4.4.2. A minimum of 75 percent of the remaining particles shall pass through a 200 mesh screen. A minimum of 95 percent of the remaining particles shall pass through a 270 mesh screen. The scrubbing pads, when tested in accordance with 4.4.5, shall remove no less than 2.00 g or no more than 6.00 g of weight when averaged from the test substrate material.

3.3.1.3 Class 3. Type I, class 3 stripping pads shall be suitable for scouring and stripping the type floors specified in 3.3.1.1 and in addition, wood and concrete floors. The abrasive content for stripping pads shall be no more than 65 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 100 mesh screen when tested in accordance with 4.4.4.2. A minimum of 99 percent of the remaining particles shall pass through a 150 mesh screen. A minimum of 85 percent of the remaining particles shall pass through a 200 mesh screen. The stripping pads, when tested in accordance with 4.4.5, shall remove no less than 3.00 g or no more than 8.00 g of weight when averaged from the test substrate material.

3.3.1.4 Class 4. Type I, class 4 polishing pads shall be suitable for polishing the type floors specified in 3.3.1.1. The abrasive content for polishing pads shall be no more than 40 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 270 mesh screen when tested in accordance with 4.4.4.2. The polishing pads, when tested in accordance with 4.4.5, shall remove no less than 0.00 g or no more than 0.25 g of weight when averaged from the test substrate material. The polishing pads shall be capable of producing a gloss reading of not less than 35 percent on the test tile when tested in accordance with 4.4.6.

3.3.2 Type II, floor pads. Type II, floor pads, classified as thin nylon floor pads, shall be in accordance with table II when tested in accordance with 4.4.3 and 4.4.3.1. The pads shall be made of a one piece, unified mat. The materials of the pads shall be formed or bonded into a solid disc. Pads of spirally wound ribbons of interlocking fibers are not acceptable.

3.3.2.1 Class 1. Type II, class 1 buffing pads shall be suitable for polishing and buffing the type floors specified in 3.3.1.1. The abrasive content for buffer pads shall be no more than 45 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 200 mesh screen when tested in accordance with 4.4.2.2. A minimum of 99 percent of the remaining particles shall pass through a 270 mesh screen. The buffing pads, when tested in accordance with 4.4.5, shall remove no less than 0.00 g or no more than 1.50 g of weight when averaged from the substrate material. The buffing pads shall be capable of producing a gloss reading of not less than 30 percent on the test tile when tested in accordance with 4.4.6.

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3.3.2.2 Class 2. Type II, class 2 scrubbing pads shall be suitable for cleaning and scrubbing the type floors specified in 3.3.1.1. The abrasive content for scrubbing pads shall be no more than 60 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 150 mesh screen when tested in accordance with 4.4.4.2. A minimum of 75 percent of the remaining particles shall pass through a 200 mesh screen. A minimum of 95 percent of the remaining particles shall pass through a 270 mesh screen. The scrubbing pads, when tested in accordance with 4.4.5, shall remove no less than 1.00 g or no more than 3.00 g of weight when averaged from the test substrate material.

3.3.2.3 Class 3. Type II, class 3 stripping pads shall be suitable for scouring and stripping the type floors specified in 3.3.1.3. The abrasive content for stripping pads shall be no more than 65 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 100 mesh screen when tested in accordance with 4.4.4.2. A minimum of 65 percent of the remaining abrasive particles shall pass through a 150 mesh screen. A minimum of 45 percent of the remaining particles shall pass through a 200 mesh screen. The stripping pads, when tested in accordance with 4.4.5, shall remove no less than 3.00 g or no more than 6.00 g of weight when averaged from the test substrate material.

3.3.3 Type III, floor pad holders.

3.3.3.1 Class 2. Type III, class 2 floor pad holders shall provide for centering, holding, and quick changing of floor pads. The holders shall be suitable for installation on new or worn bristle brushes similar to the brushes specified in 00-P-570. The bristles of the worn brushes shall be not less than 1 inch in length. The holder mounting plate shall be made of either aluminum, corrosion resisting steel or plated metal to resist corrosion from water and detergent solutions. The pad retaining collar shall be made of suitable plastic or metal disc that shall wear away without cutting or scratching floors in the event the collar contacts the floor.

3.3.3.2 Classes 3 and 4. Type III, classes 3 and 4 floor pad holders and driving heads shall provide a nonslip, quick change positive rotation of the pad capability. The positive rotation is to be accomplished without the use of mechanical devices. The driving heads may be provided with flow thru plates or shower feed capability. The class 3 material shall be "Insta-lok" type material or equal. The class 4 material shall be Velcro type fastener material or equal.

3.3.4 Type IV floor, pads. Type IV floor pads, classified as thick curled hair floor pads, shall be in accordance with table II when tested in accordance with 4.4.3 and 4.4.3.1. The pads shall be made of a one piece, unified mat. The materials of the pads shall be formed or bonded into a solid disc, and a center hole shall be provided as specified in 3.2. Pads of spirally wound ribbons of interlocking fibers are not acceptable.

3.3.4.1 Class 1. Type IV, class 1 buffing pads shall be suitable for polishing and buffing of various types of floors such as ceramic, vinyl, rubber, or asphalt tiles and marble or terrazzo. The abrasive content for buffing pads shall be no more than 35 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 200 mesh screen when tested in accordance with 4.4.4.2. A minimum of 99 percent of the remaining particles shall pass through a 270 mesh screen. The buffing pads, when tested in accordance with 4.4.5, shall remove no less than 0.00 g. or no more than 0.50 g of weight when averaged from the substrate material. The buffing pads shall be capable of producing a gloss reading of not less than 30 percent on the test tile when tested in accordance with 4.4.6.

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3.3.4.2 Class 2. Type IV, class 2 scrubbing pads shall be suitable for cleaning and scrubbing the type floors specified in 3.3.4.1. The abrasive content for scrubbing pads shall be no more than 60 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 150 mesh screen when tested in accordance with 4.4.4.2. A minimum of 75 percent of the remaining particles shall pass through a 200 mesh screen. A minimum of 95 percent of the remaining particles shall pass through a 270 mesh screen. The scrubbing pads, when tested in accordance with 4.4.5, shall remove no less than 2.00 g or no more than 6.00 g of weight when averaged from the test substrate material.

3.3.4.3 Class 3. Type IV, class 3 stripping pads shall be suitable for scouring and stripping the type floors specified in 3.3.4.1 and in addition, wood and concrete floors. The abrasive content for stripping pads shall be no more than 65 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 100 mesh screen when tested in accordance with 4.4.4.2. A minimum of 99 percent of the remaining particles shall pass through a 150 mesh screen. A minimum of 85 percent of the remaining particles shall pass through a 200 mesh screen. The stripping pads, when tested in accordance with 4.4.5, shall remove no less than 3.00 g or no more than 8.00 g of weight when averaged from the test substrate material.

3.3.4.4 Class 4. Type IV, class 4 polishing pads shall be suitable for polishing the type floors specified in 3.3.4.1. The abrasive content for polishing pads shall be no more than 40 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 270 mesh screen when tested in accordance with 4.4.4.2. The polishing pads, when tested in accordance with 4.4.5, shall remove no less than 0.00 g or no more than 0.25 g of weight when averaged from the test substrate material. The polishing pads shall be capable of producing a gloss reading of not less than 35 percent in the test tile when tested in accordance with 4.4.6.

3.3.5 Type V, floor pads. Type V floor pads, classified as thin nylon or polyester buffing floor pads, shall be in accordance with table II when tested in accordance with 4.4.3 and 4.4.3.1. The pads shall be made of a one piece, unified mat. The materials of the pads shall be formed or bonded into a solid disc. Pads of spirally wound ribbons of interlocking fibers are not acceptable. The buffing pads shall be suitable for polishing and buffing the type floors specified in 3.3.4.1. The abrasive content for buffer pads shall be no more than 40 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 200 mesh screen when tested in accordance with 4.4.4.2. A minimum of 99 percent of the remaining particles shall pass through a 270 mesh screen. The buffing pads, when tested in accordance with 4.4.5, shall remove no less than 0.20 g or no more than 3.00 g of weight when averaged from the substrate material. The buffing pads shall be capable of producing a gloss reading of not less than 30 percent in the test tile when tested in accordance with 4.4.6.

3.3.6 Type VI, floor pads. Type VI floor pads, classified as thick polyester spray pads, shall be in accordance with table II when tested in accordance with 4.4.3 and 4.4.3.1. The pads shall be made of a one piece, unified mat. The materials of the pads shall be formed or bonded into a solid disc and a center hole shall be provided as specified in 3.2. Pads of spirally wound ribbons of interlocking fibers are not acceptable.

3.3.6.1 Class 1. Type VI, class 1 spray buffing pads shall be suitable for polishing and buffing of various types of floors such as ceramic, vinyl, rubber, or asphalt tiles and marble or terrazzo. The abrasive content for spray buffing pads shall be no more than 30 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 200 mesh screen when tested in accordance with 4.4.4.2. A minimum of 99 percent of the remaining particles shall pass through a 270 mesh screen. The spray buffing pads, when tested in accordance with 4.4.5, shall remove no less than 0.10 g or no more than 1.20 g of weight when averaged from the substrate material. The buffing pads shall be capable of producing a gloss reading of not less than 30 percent on the test tile when tested in accordance with 4.4.6.

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3.3.6.2 Class 2. Type VI, class 2 spray scrubbing pads shall be suitable for cleaning and scrubbing the type floors specified in 3.3.6.1. The abrasive content for spray scrubbing pads shall be no more than 40 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 200 mesh screen when tested in accordance with 4.4.4.2. A minimum of 95 percent of the remaining particles shall pass through a 270 mesh screen. The spray scrubbing pads, when tested in accordance with 4.4.5, shall remove no less than 2.00 g or no more than 6.00 g of weight when averaged from the test substrate material.

3.3.6.3 Class 3. Type VI, class 3 spray stripping pads shall be suitable for scouring and stripping the type floors specified in 3.3.6.1 and in addition, wood and concrete floors. The abrasive content for spray stripping pads shall be no more than 55 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 100 mesh screen when tested in accordance with 4.4.4.2. A minimum of 99 percent of the remaining particles shall pass through a 150 mesh screen. A minimum of 85 percent of the remaining particles shall pass through a 200 mesh screen. The spray stripping pads, when tested in accordance with 4.4.5, shall remove no less than 2.00 g or no more than 7.00 g of weight when averaged from the test substrate material.

3.3.6.4 Class 4. Type VI, class 4 polishing pads shall be suitable for polishing the type floors specified in 3.3.6.1. The abrasive content for polishing pads shall be no more than 40 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 270 mesh screen when tested in accordance with 4.4.4.2. The polishing pads, when tested in accordance with 4.4.5, shall remove no less than 0.00 g or no more than 0.50 g of weight when averaged from the test substrate material. The polishing pads shall be capable of producing a gloss reading of not less than 35 percent on the test tile when tested in accordance with 4.4.6.

3.3.7 Type VII, floor pads. Type VII floor pads, classified as thick blended polyester and nylon pads, shall be in accordance with table II when tested in accordance with 4.4.3 and 4.4.3.1. The pads shall be made of a one piece, unified mat. The materials of the pads shall be formed or bonded into a solid disc, and a center hole shall be provided as specified in 3.2. Pads of spirally wound ribbons of interlocking fibers are not acceptable.

3.3.7.1 Class 1. Type VII, class 1 buffing pads shall be suitable for polishing and buffing of various types of floors such as ceramic, vinyl, rubber or asphalt tiles and marble or terrazzo. The abrasive content for buffing pads shall be no more than 35 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 200 mesh screen when tested in accordance with 4.4.4.2. A minimum of 99 percent of the remaining particles shall pass through a 270 mesh screen. The buffing pads, when tested in accordance with 4.4.5, shall remove no less than 0.00 g or no more than 0.50 g of weight when averaged from the substrate material. The buffing pads shall be capable of producing a gloss reading of not less than 30 percent on the test tile when tested in accordance with 4.4.6.

3.3.7.2 Class 2. Type VII, class 2 scrubbing pads shall be suitable for cleaning and scrubbing the type floors specified in 3.3.7.1. The abrasive content for scrubbing pads shall be no more than 60 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 150 mesh screen when tested in accordance with 4.4.4.2. A minimum of 75 percent of the remaining particles shall pass through a 200 mesh screen. A minimum of 95 percent of the remaining particles shall pass through a 270 mesh screen. The scrubbing pads, when tested in accordance with 4.4.5, shall remove no less than 2.00 g or no more than 6.00 g of weight when averaged from the test substrate material.

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3.3.7.3 Class 3. Type VII, class 3 stripping pads shall be suitable for scouring and stripping the type floors specified in 3.3.7.1 and in addition, wood and concrete floors. The abrasive content for stripping pads shall be no more than 65 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 100 mesh screen when tested in accordance with 4.4.4.2. A minimum of 99 percent of the remaining particles shall pass through a 150 mesh screen. A minimum of 85 percent of the remaining particles shall pass through a 200 mesh screen. The stripping pads, when tested in accordance with 4.4.5, shall remove no less than 3.00 g or no more than 8.00 g of weight when averaged from the test substrate material.

3.3.7.4 Class 4. Type VII, class 4 polishing pads shall be suitable for polishing the type floors specified in 3.3.7.1. The abrasive content for polishing pads shall be no more than 40 percent by weight when tested in accordance with 4.4.4.1. A minimum of 100 percent of the total abrasive particles shall pass through a 270 mesh screen when tested in accordance with 4.4.4.2. The polishing pads, when tested in accordance with 4.4.5, shall remove no less than 0.00 g or no more than 0.25 g of weight when averaged from the test substrate material. The polishing pads shall be capable of producing a gloss reading of not less than 35 percent on the test tile when tested in accordance with 4.4.6.

3.4 Performance. The floor maintenance pads shall wear evenly and buff, scrub, or strip uniformly as the pad wears down. The thick pads shall be capable of being washed repeatedly throughout their life without reducing their effectiveness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the contractor is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3 Visual and dimensional examination. The pad or holder shall be examined for defects in finish, construction, workmanship, marking and nonconformance with dimensional requirements. Defects shall be as classified in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 4.0 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
General	Not type specified.	X	
	Not class specified.	X	
	Not size specified.	X	
	Center hole not required size.	X	
	Thickness less than specified.	X	

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4.3.1 Functional and abrasive distribution examination. Three sample floor pads of each type and class shall be selected for the functional tests and the abrasive distribution test. Failure of any sample to pass any test shall be cause for rejection. Unless otherwise specified by the ordering activity (see 6.2), the functional and the abrasive distribution tests shall not be required if satifactorily performed on the manufacturer's product within 12 months from the date of the contract, unless there has been a major change in the manufacturing process.

4.4 Tests.

4.4.1 Test equipment. The test equipment shall consist of a Scheifer, or equal, abrasion testing machine. The Schiefer machine is designed primarily for wear testing of fabrics under "dry" conditions. The machine shall be modified for use with water or detergent solutions by providing a water resevoir, an ON-OFF stop cock, a metering stop cock, a catch basin and a drain. The upper plate shall be made to hold a 3 + 1/8 -0 inch diameter pad sample with the aid of either Velcro, Insta-Lok or similar type material. The metal bottom plate shall be 5 ± 1/16 inches in diameter. The substrate material (see 4.4.2) shall be held in place by either two small pins located near the edge and approximately 180° apart or other equally effective method to prevent slipping or turning of the substrate material during testing. A 909 g (2 pounds) head weight and a 455 g (1 pound) head weight is to be used on the Schiefer machine. Also required to accomplish the following tests are four vested sieves, 1-270 mesh with a hole size of 53 microns; 1-200 mesh screen with a hole size of 74 microns; 1-150 mesh screen with a hole size of 105 (mesh screen with a hole size of 105) microns, 1-100 mesh screen with a hole size of 149 microns and a bottom pan. Additional equipment should include a gram scale calibrated in centigrams; a leveling plate 3-1/2 inches in diameter weighing 100 g (0.2 pounds); an 85 to 100 mm diameter ashing dish; substrate material and other items as specified in the test paragraphs.

4.4.2 Substrate tile. The substrate material or test tile shall be 1/8 inch gauge, commercial, plain, solid color vinyl-asbestos tile (Armstrong No. 56807, neutral, group 11, excelon feature tile). The tile shall be cut into approximately 5 inch diameter discs. The edges shall be notched 180 degrees apart and the rough edges sanded. The discs shall be burnished for 1000 cycles or until the factory finish is removed, using a floor scrubbing pad 3 inches in diameter. The pad shall be loaded with the 1360 g (3 pounds) head weight. After burnishing, remove the substrate, wipe it clean with a soft tissue, mark an identifying number on the bottom and weigh to the nearest centigram.

4.4.3 Thickness test without load. Remove one sample floor pad types I, IV, VI, and VII (see 4.2) from shipping container and cut four 3-inch diameter discs approximately 90° apart for testing. Each cut out is conditioned to ensure that irregularity in thickness caused by storing and shipping is eliminated. Conditioning is achieved by placing the cut out on a flat, level, hard surface and allowing it to remain in that position for 48 hours at room temperature (40 to 50 percent relative humidity and 70° to 80° F temperature). After 48 hours place a flat, smooth leveling plate 3-1/2 inches in diameter, weighing 100 g (0.2 pound), concentric and horizontal on the specimen being measured. Wait 1 minute and measure distance from the flat surface to the bottom edge of the leveling plate. Take four readings, 90 degrees apart. Average the four readings and record the average obtained as the thickness of the specimen being measured. Repeat the procedure using the other three cut outs from the same pad. Average the four results. Pad thickness shall be as specified in 1.2.1 for type and class being tested.

4.4.3.1 Thickness test under load test. The same test specimens which are conditioned and tested in accordance with 4.4.3 are also used for this test. Add a 1360 g (3 pound) weight concentrically on the leveling plate. Repeat procedure used in 4.4.3. Calculate the percentage of loss in thickness of pad. This percentage shall be in accordance with table II.

After noting the results, soak the class 2 and class 3 test pads for 5 minutes in hot tap water (120°F minimum). After soaking, shake out the excess water and repeat the tests specified in 4.4.3 and 4.4.3.1. Calcualte the percentage of loss in thickness of pad. This percentage shall be in accordance with table II.

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TABLE II. Percent maximum allowable thickness loss

Type	Class	Dry	Wet
I	1	35 percent	Not applicable (N/A)
	2	25 percent	30 percent
	3	20 percent	25 percent
	4	40 percent	N/A
II	No Compression Test Required.		
IV	1	35 percent	N/A
	2	25 percent	30 percent
	3	20 percent	25 percent
	4	40 percent	N/A
V	No Compression Test Required.		
VI	1	35 percent	N/A
	2	30 percent	35 percent
	3	25 percent	30 percent
	4	45 percent	N/A
VII	1	35 percent	N/A
	2	30 percent	35 percent
	3	25 percent	30 percent
	4	45 percent	N/A

4.4.4 Ashing procedure. Heat a suitably clean ashing vessel, such as a porcelain dish, approximately 85 to 100 mm in diameter, at 600°C for 15 minutes. Cool in desiccator and weigh to nearest centigram. Add sample of the floor pad to be analyzed, favoring pieces approximately 1/2 g each, confined within the perimeter of the vessel. This technique helps to prevent some types of pad samples from floating out of the vessel. Weigh the vessel and contents. Compute and record the sample weight with the appropriate identification data. Drive off obnoxious organic volatiles under a fume hood until a dry char is obtained. Finish ignition to the ash in the muffle furnace. A furnace with regulated air ventilation is preferable because air circulation aids the ashing, which may be complete in about 15 minutes. Without air circulation, one-half to one hour may be required. When ashing is completed, remove vessel with ash to a refractory plate to cool, finishing the final cooling in a desiccator. Be careful to avoid expulsion of fluffy ash particles. When cooled to room temperature, weigh the vessel with the ash and transfer the ash to a small plastic or glass vial, properly identified and retained for the particle size distribution test (see 4.4.4.2). (NOTE: No attempt was made to remove fillers, such as clay or calcium carbonate for two reasons: (1) The procedure is complex and time consuming; and (2) most common mineral fillers act as additional abrasives and logically should be considered as part of the total abrasive structure of the pad.) Correct for calcium carbonate by measuring CO_2 loss on unashed sample.

4.4.4.1 Uniformity and abrasive content. Four 1-inch square sections shall be cut, one from each quadrant of the sample pad. The thin pads shall be tested for abrasive content only. The thick pads shall be tested for both uniformity and abrasive content. The four squares from the thin pads shall be ashed together. The residue (see 4.4.4) shall be considered the abrasive content. Each square cut from the thick pad shall be sliced into three approximately equal layers. The layers shall be identified as top, middle, and bottom layers. The four top layers shall be placed in the porcelain dish after treating (see 4.4.4), weighed to the nearest centigram, ashed and, when cooled, weighed again. Subtract the vessel weight and record the weight of the abrasive content (residue). Calculate the percentage of weight loss. Repeat the procedure for the middle layers and the bottom layers. The maximum difference of the percentages for the three layers shall be in accordance with 3.2. To find the representative total abrasive content, add the three ash weights together. Calculate the percentage of abrasive content for conformance to 3.3 as applicable.

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4.4.4.2 Abrasive distribution. Using the ash obtained in 4.4.4.1, assemble clean nested sieves with the pan, 100, 150, 200, and 230 mesh screens from top to bottom. Put the sample to be sieved into a small glass, agate or polished porcelain mortar. Rub the mass while avoiding grinding to break up agglomeration of the individual finest particles. Avoid top vigorous action that may cause static electricity on the particles; transfer the loosened mass to the top of the sieve set. Avoid shaking the sieves, because the action causes further agglomeration and could give false results favoring the larger sizes. Carefully assist the particles with crosswise, rather than downward actions on the top screen. Use small semi-stiff hair or fine bristle type brushes trimmed to about one-half inch long and other convenient tools, such as a smooth edged scraper or spatula to move the particles back and forth over the screen until all particles of appropriate sizes have passed through. Remove the top screen and repeat the procedure on the next finest screen. Proceed likewise in order from top to bottom (coarse to fine) until the finest fraction is through to the pan. Collect, weigh and record for each fraction. Total the weights of screen fractions recovered. (Note: This will invariably be a bit less than the total ash weight.) Compute the percentage of each size fraction on the basis of the total recovery weight for conformance with the paragraphs under 3.3 as applicable.

4.4.5 Abrasion test. Samples of class 1, 2, 3, and 4 floor maintenance pads, selected in accordance with 4.2, shall be tested to determine conformance to the abrasion requirements of the applicable paragraphs of 3.3. Pads used in 4.4.3 and 4.4.3.1 are acceptable for use in this test. Adjust the metering stop cock for 60 drops per minute directed onto the lower turntable of the Schiefer machine. Shut water supply off at the on-off stop-cock. A 5-inch diameter disc of substrate material shall be prepared as specified in 4.4.2, washed, dried, and weighed to the nearest centigram. Record the weight and place disc on the lower turntable. A 3-inch diameter pad shall be cut from a floor pad sample. Soak the pad in clear water (approximately 120°F) for 5 minutes.

Shake out excess water and affix to the upper turntable. (Note: Do not use water for testing the class 1 and class 4 pads. They are to be tested without water soaking or water drops.) Lower the upper turntable, start the machine and open the on-off stop cock. Rotate for 5000 revolutions. Turn off the machine and the water supply. Remove the lower tile, wash, dry with soft tissue and weigh to nearest centigram. Record weight loss and calculate percentage of initial weight. Replace tile on lower turntable. Remove pad, carefully rinse and replace with other face down and repeat above procedure for 5000 revolutions. Wash, dry, and weigh tile in centigrams for new weight loss. Record weight loss and calculate percentage of weight loss. Repeat the above procedure using two additional test pads from each sample floor pad of the same color (6 faces). Discount the highest and the lowest weight losses. Average the 4 remaining figures. The average shall be in accordance with the applicable paragraphs of 3.3. A 1360 g (3 pounds) head weight shall be used for all abrasive testing.

4.4.6 Gloss test. Samples of class 1, 4, and type V floor pads, selected in accordance with 4.2 shall be tested to determine conformance to the gloss producing requirement of 3.3 as applicable. A 5-inch diameter disc of the substrate material shall be burnished in accordance with 4.4.2. The tile shall then be covered with a brush coat of 16 percent solid content water emulsion wax conforming to Federal Specification P-W-155. The application rate shall be approximately 2000 square feet per gallon. Allow 1 hour air dry time before applying a second coat and allow another hour air dry time prior to testing. After air drying, the tile shall be placed on the Schiefer machine and polished with a 3-inch diameter pad cut from the floor pad sample. Polishing shall be accomplished by rotating the pad at a speed of 250 r.p.m. (revolutions per minute) for 200 revolutions under an applied load of a 1360 g (3-pounds) head weight. Gloss readings shall be taken with a 60° gloss meter in accordance with the procedures of test method 6101 of Fed. Test Method Std. No. 141.

4.4.7 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container, fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II, and the AQL shall be 4.5 defects per hundred units.

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TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Weight	Gross weight exceeds requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Pads of one type, class, and size shall be packaged in a box conforming to PPP-B-636, V3c. Each box shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

5.1.2 Commercial. The pads shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.2 Packing. Packaging shall be level A or Commerical as specified (see 6.2).

5.2.1 Level A. Pads of one type, class, and size shall be packed in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather resistant; or PPP-B-640, class 2, grade A. The boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification.

5.2.2 Commercial. The packaged pads shall be packed in a corrugated fiberboard box conforming to Rule 41 of the Uniform Freight Classification or to Item 222 of the National Motor Freight Classification.

5.2.3 Unitization. When shipments to Government depots are full car or truckload, the packed pads shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 lbs. in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.3 Marking. Packages, shipping containers, and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The floor pads and holders are intended for use in heavy duty floor care and maintenance.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and size required (see 1.2.1, 1.2.2.1, and 1.2.2.2).
(c) Type, class, and size required (see 3.3.3.1 and 3.3.3.2).
(d) Functional and abrasive distribution when specified (see 4.3.3).
(e) Level of packaging required (see 5.1).
(f) Level of packing required (see 5.2).

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6.3 For civil agency procurement. When level B packaging is required for civil agency procurement, the level A requirements in 5.1.1 shall apply.

6.4 Product information. It is recommended that the using agency standardize on the thick floor pads as covered by this specification. The use of "thin" pads is discouraged.

The thick pads should be cleaned immediately after use to prevent loading of the pad and to increase its life and serviceability. The pad should be cleaned with hot water, using a spray nozzle, to free dirt and other matter which has been loaded by the pad. In many instances, the pads are loaded with wax of polish particles which are very difficult to remove by water alone. In this case, the pads may be cleaned by soaking approximately one hour or more in a mild stripping solution. Pads should then be rinsed thoroughly with hot water and dried before use. (Pads may soften while in solution but original condition will return after drying.) Under no circumstances should the pads be cleaned with any type of brush which tends to destroy or loosen the fibers of the pad, as this will decrease the serviceability of the pad.

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